

# 1 Jarvis and Mckenzie, 1980 General equations

Conservation of mass, momentum and energy:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (1)$$

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \quad (2)$$

$$\rho C_p \left\{ \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right\} = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial u_{ij}} \quad (3)$$

All terms and symbols used in these equations are shown in table 1. The (Newtonian) relation between the deviatoric stress tensor  $\tau_{ij}$  and the velocity field is

$$\tau_{ij} = \eta \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) + \eta_2 \delta_{ij} \frac{\partial u_k}{\partial x_k}. \quad (4)$$

Combining 4 with eq. 3 (momentum) gives the general Navier-Stokes equations, but Jarvis and Mckenzie (1980) does not show these. Instead, they assume that  $\frac{1}{Pr} \rightarrow 0$ , which they assume (do they?) is the case for the Earth's mantle, the inertial forces can be ignored. The resulting equations will be discussed in the next section, which covers the approximations made by Jarvis and Mckenzie (1980).

## 2 Jarvis and Mckenzie, 1980 Approximate equations

Problem statement: for the Boussinesq approximation to be valid, the depth range of convection  $d$  must be small compared to the temperature scale height  $H_T$  ( $d/H_T \ll 1$ ). This is not the case for many systems of deep convection, amongst which the convection of planetary mantles. Physically,  $d/H_T \approx O(1)$  means that the influence of viscous dissipation, variable adiabatic temperature- and density gradients must be maintained in the set of governing equations.

The assumptions made for the **anelastic liquid approximation** are that  $d = O(H_T)$  and that  $\alpha T \ll 1$  what does this mean and how does this relate to Prandtl number? and if  $\alpha \ll 1$ , how come  $D$  is not  $\ll 1$ ?

## 3 Appendix 1: tables of constants

Table 1: Constants used by Jarvis and Mckenzie, 1980

| Symbol         | Value                                | Meaning                                 |
|----------------|--------------------------------------|-----------------------------------------|
| $d$            |                                      | depth of the convecting layer           |
| $H_T$          |                                      | temperature scale height of the liquid  |
| $C_p$          |                                      | specific heat at constant pressure      |
| $\alpha$       |                                      | volume coefficient of thermal expansion |
| $\tau_{ij}$    |                                      | deviatoric stress tensor                |
| $(\nabla T)_s$ | $\frac{\alpha T}{\rho C_p} \nabla P$ | adiabatic temperature gradient          |
| $K$            |                                      | thermal conductivity                    |
| $H$            |                                      | rate of internal heat production        |
| $\eta$         |                                      | dynamic viscosity                       |
| $\eta_0$       | $\ll 1$                              | second or bulk viscosity                |